WoSM Academy

MAT231-F27 · CALCULUS II

# Second-order linear equations

## Constant coefficients, oscillations and resonance

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# Last time

- Initial value problems $y' = f(t, y)$; existence and uniqueness; direction fields; Euler's method.
- Separable equations: $\int dy/h(y) = \int g(t)\,dt$.
- Linear equations $y' + py = q$: the integrating factor $e^{\int p}$; solutions are particular plus homogeneous.

Newton's law $F = ma$ involves the second derivative of position. Today: second-order linear equations with constant coefficients — springs, circuits, and anything that oscillates.

# This lecture

01 Second-order linear equations

02 The characteristic equation

03 Oscillations

04 Forced oscillations

05 Summary and practice

Reading: Strang & Herman, *Calculus, Volume 3*, Sections 7.1–7.3.

01

# Second-order linear equations

# A mass on a spring

A mass $m$ on a spring with stiffness $k$, a damper with coefficient $c \geq 0$, and an external force $F(t)$. With $x(t)$ the displacement from equilibrium, Newton's law gives

$$m\,x'' = \underbrace{-kx}_{\text{spring}} \; \underbrace{-\,c\,x'}_{\text{damping}} + F(t), \qquad \text{that is} \qquad mx'' + cx' + kx = F(t).$$

- An electric circuit with inductance $L$, resistance $R$ and capacitance $C$ gives the same equation for the charge: $Lq'' + Rq' + \frac{1}{C}q = E(t)$.
- In general: $ay'' + by' + cy = g(t)$, with constants $a \neq 0$, $b$, $c$. It is **homogeneous** if $g = 0$.

# The space of solutions

> **Theorem**
>
> For any $t_0$, $y_0$, $v_0$, the equation $ay'' + by' + cy = g(t)$ (with $g$ continuous) has exactly one solution with $y(t_0) = y_0$ and $y'(t_0) = v_0$, defined for all $t$.

Two initial conditions for a second-order equation: position and velocity.

- **Superposition**: if $y_1$, $y_2$ solve the homogeneous equation, so does $C_1y_1 + C_2y_2$.
- The homogeneous solutions form a vector space of **dimension** 2: the map $y \mapsto \big(y(0), y'(0)\big)$ is linear, and by the theorem a bijection onto $\mathbb{R}^2$.
- So if $y_1$, $y_2$ are two independent solutions, every solution is $C_1y_1 + C_2y_2$. They are independent if the **Wronskian** $W = y_1y_2' - y_1'y_2$ is not 0.

02

# The characteristic equation

# Try an exponential

For $ay'' + by' + cy = 0$, try $y = e^{rt}$:

$$ar^2 e^{rt} + bre^{rt} + ce^{rt} = (ar^2 + br + c)\, e^{rt} = 0.$$

So $e^{rt}$ is a solution exactly when $r$ solves the **characteristic equation**

$$ar^2 + br + c = 0.$$

Three cases, according to the discriminant $b^2 - 4ac$:

- $> 0$: two real roots $r_1 \neq r_2$;
- $= 0$: one double root $r$;
- $< 0$: two complex roots $\alpha \pm i\beta$.

In each case: find the roots, read off two independent solutions, and fit $C_1$, $C_2$ to the initial conditions.

# Two real roots

**Two real roots $r_1 \neq r_2$**

$$y = C_1 e^{r_1 t} + C_2 e^{r_2 t}.$$

The two solutions are independent: their Wronskian is $(r_2 - r_1)\, e^{(r_1+r_2)t} \neq 0$.

**Example** ($y'' - y' - 2y = 0$, $y(0) = 1$, $y'(0) = 2$)

$r^2 - r - 2 = (r-2)(r+1)$, so $y = C_1 e^{2t} + C_2 e^{-t}$.
$C_1 + C_2 = 1$ and $2C_1 - C_2 = 2$ give $C_1 = 1$, $C_2 = 0$: $y = e^{2t}$.

# A double root

**A double root $r = -\frac{b}{2a}$**

$$y = (C_1 + C_2 t)\, e^{rt}.$$

Why $te^{rt}$ is a solution: with $p(r) = ar^2 + br + c$, a short computation gives

$$a\,(te^{rt})'' + b\,(te^{rt})' + c\,te^{rt} = p(r)\,te^{rt} + p'(r)\,e^{rt},$$

and a double root has $p(r) = p'(r) = 0$.

**Example** ($y'' + 2y' + y = 0$)

$r^2 + 2r + 1 = (r+1)^2$: $y = (C_1 + C_2 t)\, e^{-t}$.

# Complex exponentials

The Taylor series of $e^x$ (Lecture 6) also converges for complex $x$. With $x = i\theta$, and $i^2 = -1$, the even and odd terms separate:

$$e^{i\theta} = \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots\right) + i\left(\theta - \frac{\theta^3}{3!} + \dots\right) = \cos\theta + i\sin\theta.$$

- This is **Euler's formula**. So $e^{(\alpha+i\beta)t} = e^{\alpha t}(\cos\beta t + i\sin\beta t)$, and its derivative is $(\alpha + i\beta)\, e^{(\alpha+i\beta)t}$, by the product rule.
- If $r = \alpha + i\beta$ is a root of $ar^2 + br + c$, then $e^{rt}$ is a complex solution. Since $a$, $b$, $c$ are real, its real and imaginary parts are real solutions:

$$e^{\alpha t}\cos\beta t \qquad \text{and} \qquad e^{\alpha t}\sin\beta t.$$

# Complex roots

**Two complex roots $\alpha \pm i\beta$ ($\beta \neq 0$)**

$y = e^{\alpha t}\left(C_1 \cos \beta t + C_2 \sin \beta t\right).$

**Example** ($y'' + 4y = 0$, $y(0) = 1$, $y'(0) = 2$)

$r = \pm 2i$: $y = C_1 \cos 2t + C_2 \sin 2t$. $y(0) = C_1 = 1$, $y'(0) = 2C_2 = 2$: $y = \cos 2t + \sin 2t$.

**Example** ($y'' + 2y' + 5y = 0$: **an oscillation that dies out**)

$r = -1 \pm 2i$: $y = e^{-t}\left(C_1 \cos 2t + C_2 \sin 2t\right).$

03

# Oscillations

# Free undamped oscillation

$mx'' + kx = 0$, that is $x'' + \omega_0^2 x = 0$ with $\omega_0 = \sqrt{k/m}$. Roots $\pm i\omega_0$:

$$x(t) = C_1 \cos \omega_0 t + C_2 \sin \omega_0 t = A \cos(\omega_0 t - \delta),$$

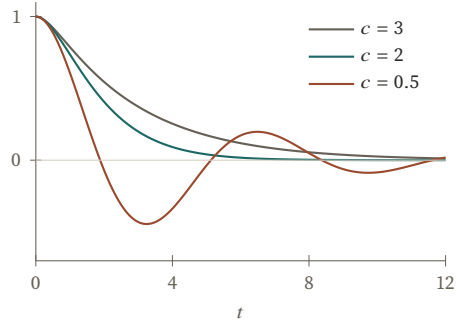
with **amplitude** $A = \sqrt{C_1^2 + C_2^2}$ and phase $\delta$ ($\cos \delta = C_1/A$, $\sin \delta = C_2/A$).

- Simple harmonic motion with **natural frequency** $\omega_0$ and period $2\pi/\omega_0$.
- A stiffer spring (larger $k$) or lighter mass (smaller $m$) oscillates faster.
- The energy $\frac{1}{2}m(x')^2 + \frac{1}{2}kx^2$ is constant: differentiate it and use the equation.

# Damping

$mx'' + cx' + kx = 0$ with $c > 0$: roots $\dfrac{-c \pm \sqrt{c^2 - 4mk}}{2m}$, both with negative real part.

- $c^2 > 4mk$: **overdamped**. Two negative real roots: a slow return, no oscillation.
- $c^2 = 4mk$: **critically damped**: the fastest return without overshoot.
- $c^2 < 4mk$: **underdamped**. A decaying oscillation $e^{-ct/2m}\cos(\mu t - \delta)$.




$m = k = 1,\ x(0) = 1, x'(0) = 0.$

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# Forced oscillations

# Nonhomogeneous equations

**Theorem**

The general solution of $ay'' + by' + cy = g(t)$ is $y = y_p + C_1 y_1 + C_2 y_2$: one particular solution plus the general homogeneous solution. (The difference of two solutions solves the homogeneous equation.)

**Undetermined coefficients**: guess $y_p$ of the same form as $g$.

| $g(t)$ | try $y_p$ |
|---|---|
| polynomial of degree $n$ | polynomial of degree $n$ |
| $e^{kt}$ | $Ae^{kt}$ |
| $\cos \omega t$ or $\sin \omega t$ | $A \cos \omega t + B \sin \omega t$ |

If the guess solves the homogeneous equation, multiply it by $t$ (again, if necessary).

# Examples

**Example** ($y'' - y' - 2y = 4t$)

Try $y_p = At + B$: $0 - A - 2(At + B) = 4t$, so $A = -2$ and $B = 1$. With the homogeneous solutions from before:

$$y = C_1 e^{2t} + C_2 e^{-t} - 2t + 1.$$

**Example** ($y'' + 4y = 3\cos t$)

Try $y_p = A\cos t$ (no $\sin t$ is needed, as there is no $y'$ term): $-A + 4A = 3$, so $A = 1$:

$$y = \cos t + C_1 \cos 2t + C_2 \sin 2t.$$

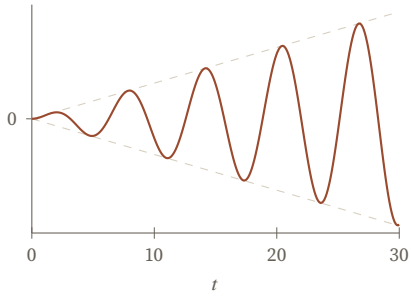
# Resonance

$x'' + \omega_0^2 x = \cos \omega t$: a spring pushed at frequency $\omega$.

- $\omega \neq \omega_0$: $x_p = \dfrac{\cos \omega t}{\omega_0^2 - \omega^2}$, which grows as $\omega \to \omega_0$.
- $\omega = \omega_0$: the guess $A \cos \omega_0 t$ solves the homogeneous equation; multiplying by $t$ gives

$$x_p = \frac{t \sin \omega_0 t}{2\omega_0},$$

which grows without bound: **resonance**.



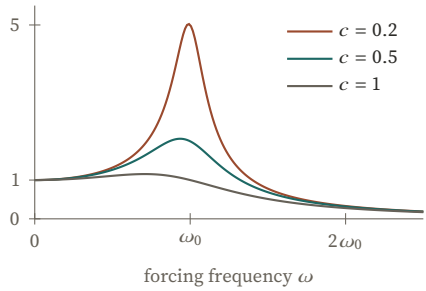

$x'' + x = \cos t,\ x_p = \frac{1}{2} t \sin t.$

# Damping keeps the response bounded

$mx'' + cx' + kx = \cos \omega t$ with $c > 0$ has a solution $A \cos \omega t + B \sin \omega t$ of amplitude

$$\frac{1}{\sqrt{(k - m\omega^2)^2 + c^2\omega^2}}.$$

- The homogeneous part decays: every solution approaches this **steady state**.
- It peaks near $\omega_0 = \sqrt{k/m}$, higher for smaller $c$.




Amplitude, $m = k = 1$.

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# Summary and practice

# Summary

- $ay'' + by' + cy = 0$: solutions form a 2-dimensional space; two initial conditions fix one.
- Characteristic equation $ar^2 + br + c = 0$. Real roots: $C_1 e^{r_1 t} + C_2 e^{r_2 t}$; double root: $(C_1 + C_2 t)e^{rt}$; complex roots $\alpha \pm i\beta$: $e^{\alpha t}(C_1 \cos \beta t + C_2 \sin \beta t)$, by Euler's formula.
- Springs: simple harmonic motion; over-, critically and underdamped motion.
- Forced: $y = y_p + y_h$; undetermined coefficients; resonance when the forcing matches the natural frequency.

# Practice

1. Solve $y'' - 5y' + 6y = 0$, $y(0) = 1$, $y'(0) = 0$.
2. Solve $y'' - 4y' + 4y = 0$, $y(0) = 1$, $y'(0) = 0$.
3. Find the general solution of $y'' + 2y' + 10y = 0$.
4. A mass $m = 1$ hangs on a spring with $k = 4$. For which damping coefficients $c$ is the motion underdamped, critically damped, overdamped?
5. Find the general solution of $y'' + y' - 2y = 2e^{3t}$.
6. Find a particular solution of $y'' + 9y = \sin 3t$.
7. Show that the Wronskian of $e^{r_1 t}$ and $e^{r_2 t}$ is $(r_2 - r_1)\, e^{(r_1 + r_2)t}$.

# Answers

1. $r = 2, 3$: $C_1 + C_2 = 1$, $2C_1 + 3C_2 = 0$: $y = 3e^{2t} - 2e^{3t}$.
2. $r = 2$ (double): $y = (1 + C_2 t)e^{2t}$, $y'(0) = 2 + C_2 = 0$: $y = (1 - 2t)\,e^{2t}$.
3. $r = -1 \pm 3i$: $y = e^{-t}(C_1 \cos 3t + C_2 \sin 3t)$.
4. $c^2 - 16$: underdamped for $c < 4$, critical for $c = 4$, overdamped for $c > 4$.
5. $r = 1, -2$; $y_p = Ae^{3t}$ with $(9 + 3 - 2)A = 2$: $y = C_1 e^{t} + C_2 e^{-2t} + \frac{1}{5}e^{3t}$.
6. Resonance: try $t(A \cos 3t + B \sin 3t)$; $y_p = -\frac{t \cos 3t}{6}$.
7. $W = e^{r_1 t} \cdot r_2 e^{r_2 t} - r_1 e^{r_1 t} \cdot e^{r_2 t} = (r_2 - r_1)\,e^{(r_1 + r_2)t}$.

# Next time

**Lecture 25 — Systems of linear differential equations** (Unit 25)

- $\mathbf{x}' = A\mathbf{x}$: solutions from eigenvalues and eigenvectors
- Phase portraits: saddles, nodes, spirals; stability
- The matrix exponential, and a look back at the module

Reading: Apostol, *Calculus, Volume 2*, Chapter 7.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.